

[19] Secure Hash Standard (SHS)

- **Authors:** National Institute of Standards and Technology (NIST)
 - **Venue & Year:** Federal Information Processing Standards Publication (FIPS PUB 180-4, 2015)
 - **Category:** Cryptographic Content Hashing
 - **Associated Full Paper PDF:** [nist_2015_sha256_fips180-4.pdf](#)
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1-Minute Executive Takeaway

Key Finding: The official standard for SHA-256, providing the cryptographic foundation for Kwiz's 549x cache speedup.

Core Research Objective

To specify secure hash algorithms (SHA-1, SHA-224, SHA-256, SHA-384, SHA-512) for computing a condensed representation of electronic data.

Methodology & Architecture

Defined mathematical bitwise operations, padding constants, and compression functions guaranteeing collision resistance.

Key Empirical Findings & Evidence

SHA-256 generates a deterministic 256-bit cryptographic message digest with negligible collision probability, making it the industry standard for content integrity verification.

Direct Relevance to Our Kwiz Paper

Cited in Section 3.1 and Section 4.5. Kwiz computes SHA-256 hashes for each normalized course chunk, using the hash as an immutable cache lookup key to detect unchanged syllabus content and bypass redundant embedding computation.